

Herald



Herald — Continuation

Field	Value
Revision	18
Created	2026-05-20
Last modified	2026-05-28
Status	active
Status summary	2026-05-28 (day-2) — v1.0.0 Batch B COMPLETE + constitution sync + 3 fixes (HEAD 74649bb; constitution pulled 3c9c4e9→15cd4bc). (a) Constitution sync — pulled HelixConstitution 3c9c4e9→15cd4bc (gained §11.4.95 workable-items-DB-tracked-not-gitignored / §11.4.96 safe-parallel-work-with-long-build catalogue / §11.4.97 max-idle + progress-update cadence) and propagated all three into CLAUDE.md r13 / AGENTS.md r9 / HERALD_CONSTITUTION.md r7 §108.j-l (commit d5b9895); Herald already complied with §11.4.95 (DB version-controlled per the 2026-05-27 operator divergence); inheritance gate 15 PASS / 0 FAIL. (b) Three real fixes — #194 audit_stress_chaos_test.go now gates its qaRoot on HERALD_STRESS_QA_DIR via t.TempDir() (commit a8ad4f7, stops the suite littering docs/qa/ on every run); commons_constitution/eventbus.go PublishedByType data race → sync.Map.LoadOrStore (commit f5d1367 — a REAL -race regression on E3 latent since Batch A, surfaced by the Batch-B bindings stress test); E17 e2e SKIP-guard now requires a live Postgres (commit 75e7a49 — E17 was FAILing instead of honestly SKIPping when PG is absent). (c) v1.0.0 Batch B COMPLETE — all 7 §42 flavor constitution bindings (HRD-019..025) landed, wiring each flavor to emit through the Batch-A commons_constitution Evaluator+audit foundation: HRD-019 cherald (f33b5e4, e2e E90 c3e3cff), HRD-020 sherald (5ebad4f, E91 75e7a49), HRD-021 bherald (46ca77e, E92 f31615f), HRD-022 rherald (f73195f, E93 0206f8f), HRD-023/024/025 pherald/iherald/scherald (3fb753a, landed in one parallel pass; E94/E95/E96 74649bb). Each flavor: <flavor>/internal/bindings/ Pipeline + rule catalogue + bespoke PURE detectors, -race -count=3 green, §107.x evidence under docs/qa/HRD-0XX-*/ , §11.4.85 stress (1200 concurrent → 1200 emits+audit rows). All HRD-019..025 remain in_progress — mechanism + rules + detectors are done, but live operation-interception (the Subjects fed to the detectors) is Batch C §43 command bodies. e2e invariants now run through E96 (E89 Batch-A foundation + E90-E96 Batch B). Prior r13 (still valid history): 2026-05-27 v1.0.0 Batch A

Field**Value**

complete checkpoint (HEAD 854d301; constitution at 3c9c4e9). v1.0.0 readiness workstream (workstream 3) STARTED — plan docs/superpowers/plans/2026-05-27-v1.0.0-execution.md (9da90ba), 39 HRDs + Wave 7 batched A→E with tag cadence v0.5→v1.0.0. Batch A DONE (commits b19cb43 + 854d301): HRD-018 / HRD-026 / HRD-027 closed atomically Issues→Fixed (Issues.md r16→r17, 40→37 open). Closed the commons_constitution constitution_audit write-through §107 PASS-bluff — the table existed in migration 000006 but the Runner never wrote a row (RunOutcome.Audited=true persisting 0 rows); FIXED with an AuditStore interface + state.MemoryAudit + state.PostgresAudit (RLS-guarded append-only INSERT) + a new IDemitter (emit returns the generated event ID) so Runner.Run writes a durable audit row for every CHANGED Warn/Enforce transition carrying old/new decision+digest, bundle_hash, mode, and the EXACT emitted event ID for Enforce. HRD-026 bundle-hash now persisted on every emitted event AND audit row (replay §42.1.3). HRD-027 mode-ladder runtime config wired to 3 JWT-gated admin REST endpoints on cherald (GET/GET-one/PUT / v1/compliance/modes[:rule_id], flip allow/warn/enforce per tenant, no redeploy). Conductor-verified: -race -count=3 deterministic, real-PG integration green, full scripts/e2e_bluff_hunt.sh 81 PASS / 0 FAIL (was 78; +3 = E89 wired). Evidence docs/qa/HRD-018-20260527T142556Z/ (real-PG emit→persist + audit DB-level SELECT proof + admin-REST flip round-trip + §11.4.85 stress 512 concurrent emit→persist = exactly 512 audit rows no-lost-writes + chaos 960 concurrent flips). §11.4.93 SQLite-SSoT workstream 2 DEFERRED by operator (2026-05-27 decision): the constitution's workable-items migration Go binary (constitution/scripts/workable-items/) is an unimplemented scaffold (0/7 subcommands) AND schema-incompatible with Herald's HRD-NNN pipe-table format; per §11.4.74 catalogue-first Herald can NOT reimplement it — the fix is a constitution-repo task affecting all consumers. Assessment docs/research/workable-items-phase2-assessment-2026-05-27.md (36fa90b). HRD-131 stays OPEN, blocked-on-constitution-binary. Prior r12 (still valid history): **2026-05-27 GAP-3 + HRD-132 close-out checkpoint (HEAD 4f45f26; constitution at 3c9c4e9). GAP-3 §11.4.85 stress+chaos suite (HRD-122..HRD-130) AND the HRD-132 Runner idempotency-window fix are COMPLETE and now atomically migrated Issues→Fixed (Issues.md r16 / Fixed.md r15, §11.4.19): HRD-122..HRD-130 Task→Completed, HRD-132 Bug→Fixed. HRD-132 fix 354b883 lands claim-before-dispatch at Runner Stage 2 — INSERT events_processed ... ON CONFLICT DO NOTHING is the AUTHORITATIVE exactly-once dispatch gate (the Stage-2→Stage-7 duplicate-dispatch window HRD-125 surfaced is closed); latent runner.go:132 CachedRcpt.WasReplay race fixed; HRD-125 assertion upgraded to sends==1 (docs/qa/HRD-125-stress-chaos-2026-05-27T131251-gap3/ shows dispatch_exactly_once=1, shared_key_sends=1 want=1); wave3 mutation-gate M4 retired (4f45f26) — Stage-7 Insert now redundant. Full scripts/e2e_bluff_hunt.sh 78 PASS / 0 FAIL; HRD-130 gate 6 PASS / 0 FAIL (5 load-bearing mutations). HRD-131 (SQLite SSoT migration Phase 1) remains OPEN; Phases 2-6 follow-on. Prior GAP-3 checkpoint (HEAD d2a60d3, all 4 mirrors synced): GAP-3 §11.4.85 stress+chaos suite COMPLETE (commits 02d272d→d2a60d3, HRD-122..130): HRD-122 scaffold commons/stresschaos/, HRD-123 Gin /v1/events, HRD-124 /v1/

Field	Value
	compliance+/v1/safety_state, HRD-125 Runner, HRD-126 pherald listen, HRD-127 claude_code (testdata fake-claude shims), HRD-128 container/resource-exhaustion (§12.6-safe, live OOM SKIP-with-reason), HRD-129 e2e E81-E88, HRD-130 paired §1.1 mutation gate tests/ test_stress_chaos_mutation_meta.sh. Full scripts/ e2e_bluff_hunt.sh now reports 78 PASS / 0 FAIL (was 60; +18 = E81-E88 + anchor sub-checks); HRD-130 gate 6 PASS / 0 FAIL (5 mutations load-bearing). Evidence committed under docs/qa/HRD-12{3,4,5,7,8}-*+docs/qa/2026-05-27T135301-gap3-listen-HRD-126/. NOTE — HRD-122..130 are now MIGRATED to Fixed.md (Completed) (suite landed + e2e green + atomic Issues→Fixed migration done in Issues.md r16 / Fixed.md r15). Two anti-bluff issues caught + fixed mid-session by the conductor: HRD-123 flaky oversized-body assertion (§11.4.50 — transport-rejection of an 8 MiB body is valid, not a failure) and E82 stale-anchor (re-captured committed evidence with HERALD_STRESS_QA_DIR; assertion all_4xx_no_panic→all_malformed_rejected_no_5xx). HRD-132 (FIXED 354b883, now in Fixed.md): Runner idempotency window — concurrent same-key replay duplicate-dispatched (2-4 of a 1000× flood) because events_processed was archived at Stage 7 while idempotency.go:64-72 treated SETNX-miss+no-PG-row as fresh (documented §32.1 "Redis-lies-PG-truths"). FIX landed claim-before-dispatch (INSERT events_processed ... ON CONFLICT DO NOTHING at Stage 2 = authoritative exactly-once dispatch gate) + closed the latent runner.go:132 CachedRcpt.WasReplay race + upgraded the HRD-125 assertion to sends==1 (evidence dispatch_exactly_once=1); wave3 M4 retired (4f45f26). Prior r10 (still valid history) — a constitution-compliance + SSoT-hygiene sweep: (1) pulled HelixConstitution 3a085b9→3c9c4e9 (gained §11.4.89 background-test / §11.4.90 Obsolete-status / §11.4.91 summary-clarity / §11.4.92 multi-pass-eval / §11.4.93 SQLite-SSoT / §11.4.94 zero-idle-parallel); (2) propagated §11.4.89-94 into CLAUDE.md r12 / AGENTS.md r8 / HERALD_CONSTITUTION.md r6 §108.d-i (commit 83b5e2f) — inheritance gate 15 PASS / 0 FAIL; (3) built scripts/mutation_residue_audit.sh §107.y pre-push residue scanner (e23933c) + back-filled §107.y quiescence + .git/MUTATION_IN_PROGRESS lockfile into wave2/3/4 gates (c19a092); (4) re-verified ALL mutation gates green one-at-a-time — wave2 5/0, wave3 6/0, wave4 5/0, wave4b 5/0, wave6.5 6/0 (wave6 already 4/4), each with Quiescence assertion PASS; (5) §11.4.91 summary-clarity — 7 one-liners fixed (d24e829); (6) operator correction — docs/.workable_items.db is VERSION-CONTROLLED, NOT gitignored (Herald divergence from parent §11.4.93, 30f9c6d); (7) GAP-3 §11.4.85 stress+chaos PLAN committed (a422c53, HRD-122..130); (8) read-only audits committed (9c09496 constitution-compliance, ee63ce6 HRD-obsolescence + qa-coverage); (9) HRD-131 filed (SQLite SSoT Phase 1) in Issues.md. RESOLVED debts: Wave 6.5 mutation-gate M3 stale anchor (opts.ReplyTo→textOpts.ReplyTo) FIXED; Postgres-SASL e2e carry-over RESOLVED (e2e now 60 PASS / 0 FAIL / 24 SKIP); HRD-101 now filed as a real Fixed.md row. Prior session: v0.4.0 (Wave 6 closed-loop) + v0.5.0 (Wave 6.5 lifecycle) tagged; Wave 7 T1-T5 closed (HRD-110..114). PENDING-OPERATOR: MTPROTO real-channel automation (Telegram never relays bot-to-bot in groups — docs/research/telegram-bot-to-bot-constraint.md, 5267f14; needs my.telegram.org app_id+app_hash + / revoke @pherald_qa_bot).

Field	Value
	HRD-008, HRD-015, HRD-019..HRD-025 (Batch B bindings landed — in_progress , live-op deferred to Batch C), HRD-029..HRD-056, HRD-081, HRD-085..HRD-090, HRD-115..HRD-121 (Wave 7 T6-T12 pending), HRD-131 (SQLite SSoT Phase 1, OPEN-deferred — blocked on constitution binary) — 37 open
Issues summary	see <code>Issues.md</code> . v1.0.0 Batch A migrated HRD-018 / HRD-026 / HRD-027 Issues→Fixed (§11.4.19 atomic). v1.0.0 Batch B (HRD-019..025, §42 flavor constitution bindings) is in_progress — each flavor's <code>internal/bindings/Pipeline + rule catalogue + PURE detectors + emit-through-audit foundation + §107.x evidence + §11.4.85 stress</code> all landed (<code>f33b5e4/5ebad4f/46ca77e/f73195f/3fb753a</code>), but live operation-interception (the Subjects the detectors classify) is deferred to Batch C §43 command bodies, so these stay open not Fixed. Wave 7 closed HRD-110/111/112/113/114; HRD-115..121 are the remaining Wave 7 tasks T6-T12. HRD-122..HRD-130 (§11.4.85 stress+chaos suite, GAP-3) + HRD-132 (Runner idempotency-window bug) MIGRATED to Fixed.md. HRD-131 is §11.4.93 SQLite workable-items SSoT migration Phase 1 — OPEN-deferred per operator 2026-05-27: Phase 2 adoption blocked on the unimplemented + schema-incompatible constitution migration binary (assessment <code>36fa90b</code>); a §11.4.74 constitution-repo task, not a Herald reimplementation.
Fixed	HRD-001..HRD-018 (less open), HRD-026, HRD-027, HRD-028, HRD-080, HRD-092..HRD-101, HRD-110..HRD-114, HRD-122..HRD-130, HRD-132
Fixed summary	see <code>Fixed.md</code>
Continuation	Resume — v1.0.0 readiness (workstream 3) per plan docs/superpowers/plans/2026-05-27-v1.0.0-execution.md; Batch A + Batch B DONE, next is Batch C: (1) Batch C — §43 command catalogue HRD-029..056 (IMMEDIATE NEXT UNIT) — the live CLI command bodies that PRODUCE the Subjects the Batch-B detectors classify (pherald commit-push, sherald destructive-guard, cherald creds-scan, rherald tag-mirror, bherald evidence-capture, etc.), grouped into 5 flavor clusters per the v1.0.0 plan. Closing Batch C is what flips HRD-019..025 from <code>in_progress</code> →Fixed (their bindings have no live Subjects until the command surface lands). NOTE — the e2e invariant counter is at E96; next free is E97+. (2) Batch D — commons_infra HRD-081 + HRD-085..090. (3) Batch E — Wave 7 T6-T12 Slack adapter HRD-115..121 (vendored slack-go submodule — verify the I6 inheritance-gate check after the .gitmodules entry ; confirm slack-go v0.16.0 field names <code>inner.Files/socketmode.EventTypeEventsAPI/UploadFileV2Parameters</code> ; then <code>qaherald Slack MessengerClient</code> , spec §11.0/§32.2/§43, e2e, mutation gate, Issues→Fixed/v0.6.0/4-mirror push; T5 pre-staged <code>perChannelConfig("slack") + blank-import in listen.go — HERALD_CHANNELS=tgram, slack currently errors ErrUnknownChannel</code> until T6 registers slack, by design; plan docs/superpowers/plans/2026-05-27-wave7-generic-messenger.md line 980). PLAN GOTCHAS: Wave 7 e2e invariants MUST renumber off E81-E89 — use E90+ (E89 is now consumed by Batch A's audit write-through); the slack-go submodule addition requires re-running the I6 inheritance-gate check (<code>tests/test_constitution_inheritance.sh</code>). Tag cadence: v0.5→v1.0.0 across the batches. (5) GAP-4 docs/qa back-fill + docs-audit #147 — features lacking docs/qa/<run-id>/evidence per the <code>ee63ce6</code> qa-coverage audit + comprehensive documentation audit. DEFERRED / GATED ITEMS: §11.4.93 SQLite SSoT migration (HRD-131) — OPERATOR-DEFERRED 2026-05-27: Phase 2 adoption is blocked on the constitution's workable-items migration binary,

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	which is an unimplemented scaffold (0/7 subcommands) + schema-incompatible with Herald's HRD-NNN pipe-table format; per §11.4.74 Herald may NOT reimplement it — the fix is a constitution-repo task affecting all consumers (assessment docs/research/workable-items-phase2-assessment-2026-05-27.md, 36fa90b). HRD-131 stays OPEN, blocked-on-constitution-binary. MTPProto real-channel automation — OPERATOR-DECISION: provide my.telegram.org app_id+app_hash (folds into a qaherald MessengerClient MTPProto impl reusing the entire scenario engine; Telegram never relays bot-to-bot in groups — see docs/research/telegram-bot-to-bot-constraint.md); operator should also /revoke @pherald_qa_bot (leaked plaintext token). §11.4.90 obsolescence — zero current Obsolete rows; convention in place, periodic re-audit pending. (RESOLVED this session (2026-05-28 day-2): v1.0.0 Batch B COMPLETE — all 7 §42 flavor constitution bindings HRD-019..025 landed (f33b5e4/5ebad4f/46ca77e/f73195f/3fb753a), each internal/bindings/ Pipeline + rule catalogue + PURE detectors + §107.x evidence + §11.4.85 stress 1200-concurrent, e2e E90-E96 wired (HRD-019..025 stay in_progress — live-op interception is Batch C); constitution pulled 3c9c4e9→15cd4bc + §11.4.95-97 propagated (d5b9895, gate 15/0); 3 fixes — #194 audit-test qaRoot t.TempDir guard (a8ad4f7), eventbus PublishedByType data race → LoadOrStore (f5d1367, a real E3 -race regression latent since Batch A), E17 e2e SKIP-guard now requires live PG (75e7a49). Prior r13: v1.0.0 Batch A — HRD-018/026/027 closed atomically Issues→Fixed, e2e 81 PASS / 0 FAIL (E89 wired), commons_constitution audit write-through §107 bluff FIXED + admin mode-flip REST; conductor-verified -race -count=3 + real-PG + §11.4.85 stress 512-row no-lost-writes. Earlier: GAP-3 §11.4.85 stress+chaos suite IMPLEMENTED — e2e 78 PASS / 0 FAIL — AND atomically migrated Issues→Fixed (HRD-122..130 Completed); HRD-132 Runner idempotency-window bug FIXED 354b883 (claim-before-dispatch, exactly-once sends==1) + migrated to Fixed.md; constitution §11.4.89-94 propagation + residue scanner; HRD-123 flaky oversized-body assertion + E82 stale-anchor anti-bluff issues fixed. Prior r10 RESOLVED: Wave 6.5 mutation-gate M3 stale anchor; Postgres-SASL e2e FAILs; HRD-101 filed as a real Fixed.md row.) §11.4.87 endless-loop: continue until Issues.md zero-active + CONTINUATION §3 empty + no subagent in flight. NO BLUFF. Push 4 mirrors every 2-3 commits.

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§0. How to use this document

Paste the following block into any CLI agent (Claude Code / OpenCode / Cursor / Aider / Gemini CLI) to resume Herald work exactly where it was left:

You are working on the Herald project at ~/Projects/Herald (also reachable as the Herald/ submodule of a consuming project). The Helix Universal Constitution lives at <ancestor>/constitution/ (parent-walk discovery). Read in this order: CLAUDE.md, AGENTS.md, README.md, docs/guides/HERALD_CONSTITUTION.md, docs/guides/CONSTITUTION_INHERITANCE.md, docs/specs/mvp/specification.V3.md. Then read docs/CONTINUATION.md (this file) for live state, docs/Issues.md for open work, docs/Status.md for current phase, docs/Fixed.md for closed history. Go workspace builds via go test ./commons/... ./commons_prefix/... ./commons_messaging/... ./commons_storage/.... Inheritance gate tests/test_constitution_inheritance.sh MUST exit 0 before any commit. Multi-mirror fan-out push to four hosts (GitHub + GitLab + GitFlic + GitVerse) is mandatory per Constitution §103.

§1. Snapshot

- **Active spec:** docs/specs/mvp/specification.V3.md Revision 4 (~4300 lines).
- **Archived specs:** V1 + V2 in docs/specs/mvp/archive/ (frozen).
- **Go modules:** commons, commons_prefix, commons_messaging, commons_storage, pherald (all compile + unit tests pass).
- **Container scaffold:** quickstart/{docker-compose.quickstart.yml, Dockerfile.pherald, otel-config.yaml, .env.example} for §26.5 quickstart.
- **CLI:** pherald version --json returns canonical build info; serve/send/doctor/migrate/subscriber/deadletter stubbed with HRD-NNN pointers.
- **Inheritance gate:** 15 PASS / 0 FAIL. Meta-test ✓.
- **e2e bluff hunt:** invariants now run through **E120** (E89 Batch-A; E90-E96 Batch-B bindings; **E97-E120 = Batch C §43 command catalogue** — E97-E100 pherald, E101-E104 sherald, E105-E107 rherald, E108-E109 bherald, E110-E120 cherald; prior E81-E88 GAP-3 stress+chaos). Full e2e CONFIRMED 2026-05-28 = **128 PASS / 0 FAIL** (E3 -race spans 12 packages incl commons/gitops; E17 SKIP-with-reason on absent live-PG; E53 flake root-caused + fixed — was a cross-call byte-diff on the dynamic /v1/safety_state body). Plus SKIP-with-reason for hardware/credential-absent invariants (live OOM, MTProto, Wave-6 live-loop). Next free invariant id is **E121+**.
- **Stress+chaos mutation gate:** tests/test_stress_chaos_mutation_meta.sh (HRD-130) 6 PASS / 0 FAIL (5 load-bearing mutations).
- **Phase:** implementation rl.

§2. Last commit landed

This commit (V3 r4 + Go scaffold + tracking-doc refresh) closes HRD-009/HRD-009b/HRD-013/HRD-014 (with a Universal §11.4.19 atomic Issues.md → Fixed.md migration) and lands spec V3 §37–§41 (tracker events, workable-item announcement contract, message presentation + Herald Canonical Template, docs/tests completeness, REST API surface). Builds + tests:

```
$ go test ./commons/... ./commons_prefix/... ./commons_messaging/... ./commons_storage/...
ok      github.com/vasic-digital/herald/commons 1.135s
ok      github.com/vasic-digital/herald/commons_prefix 0.639s
ok      github.com/vasic-digital/herald/commons_messaging/channels/null 0.
ok      github.com/vasic-digital/herald/commons_messaging/dispatch/claude_
ok      github.com/vasic-digital/herald/commons_storage 1.630s
```

§3. Active work

ID	Status	What
HRD-008	in_progress	Operator-side Quickstart compose validation (Postgres + Redis + OTel + pherald container) — scaffold shipped this commit; live end-to-end run pending operator.
HRD-010	open	commons_storage live (pgx pool + golang-migrate driver + River queue + Redis ACL).
HRD-011	open	Telegram adapter live (telebot SDK + getUpdates long-poll + webhook secret_token).
HRD-012	open	Claude Code dispatcher live (claude --resume + parse <<<HERALD-REPLY>>>).
HRD-015	open	Inheritance gate I8 invariants for Go scaffold (go.work + commons/types.go + null adapter test passes).
HRD-016	open	REST API surface via Gin Gonic per spec §41.
HRD-017	open	Propagate Universal §11.4.6X spec-versioning + submodule-catalogue-first mandates into the parent constitution.
HRD-122..130	Completed	GAP-3 §11.4.85 stress+chaos suite — IMPLEMENTED + e2e-green (d2a60d3); MIGRATED Issues→Fixed (Fixed.md r15). DONE.
HRD-132	Fixed	Runner idempotency-window bug — FIXED 354b883 (claim-before-dispatch at Stage 2, exactly-once sends==1, CachedRcpt.WasReplay race closed); MIGRATED Issues→Fixed (Fixed.md r15). DONE.
HRD-018/026/027	Fixed	v1.0.0 Batch A (b19cb43+854d301): commons_constitution audit write-through (closed the §107 PASS-bluff — table existed, no row written) via AuditStore+Memory/Postgres backends + IDemitter; bundle-hash persisted on every event+audit row (026); cherald admin mode-flip REST GET/GET-one/PUT /v1/compliance/modes[:rule_id] (027). e2e 78→ 81/0 (E89). MIGRATED Issues→Fixed (Issues.md r17, 40→37 open). DONE.
HRD-131	OPEN-deferred	§11.4.93 SQLite workable-items SSoT migration Phase 1 filed; Phase 2 OPERATOR-DEFERRED 2026-05-27 — blocked on the constitution's migration binary (0/7 subcommands + schema-incompatible; §11.4.74 constitution-repo task, not a Herald reimplementation; assessment 36fa90b).
HRD-019..025	in_progress	v1.0.0 Batch B COMPLETE (bindings) — all 7 §42 flavor constitution bindings landed: HRD-019 cherald (f33b5e4, E90 c3e3cff), HRD-020 sherald (5ebad4f, E91 75e7a49), HRD-021 bherald (46ca77e, E92 f31615f), HRD-022 rherald (f73195f, E93 0206f8f), HRD-023/024/025 pherald/iherald/scherald (3fb753a, E94/E95/E96 74649bb). Each: <flavor>/internal/bindings/ Pipeline + rule catalogue + PURE detectors, emit-through Batch-A audit foundation, -race -count=3 green, §107.x

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#194 / eventbus race / E17	Fixed	evidence docs/qa/HRD-0XX-*/ , §11.4.85 stress 1200-concurrent → 1200 emits+audit rows. Stay in_progress — live operation-interception (the Subjects the detectors classify) is Batch C §43 command bodies; these flip →Fixed once Batch C lands. 3 day-2 fixes: #194 audit_stress_chaos_test.go qaRoot gated on HERALD_STRESS_QA_DIR via t.TempDir() (a8ad4f7, stops docs/qa noise); commons_constitution/eventbus.go PublishedByType data race → sync.Map.LoadOrStore(f5d1367, real E3 -race regression latent since Batch A); E17 e2e SKIP-guard now requires live PG (75e7a49, was FAILing instead of SKIPping when PG absent). §43 command catalogue — all 5 flavor clusters landed, each a live CLI command feeding its Batch-B binding (DETECT+classify+exit-non-zero): C1 pherald commit-push/submodule-propagate/install-upstreams/fetch-guard/reopen/pre-push (502ebf6,65c8362, E97-E100); C2 sherald destructive-guard/constitution-pull/force-push-gate/mem-budget-watch (ed6c675, E31+E101-E104); C4 rherald tag-mirror/changelog-generate/gate-retest + C5 bherald test-tier-verify/evidence-capture (130ec19, E105-E109); C3 cherald 11 docs/compliance cmds (aef0207, E110-E120). Remaining §43: HRD-034 (sherald backup-snapshot §9.3) + HRD-047 (scherald status-digest §11.4.45/.56) — not yet implemented. Rows are in_progress (hermetic+e2e proven; flip →Fixed at an operator-live close-out pass, together with HRD-019..025).
Batch C (HRD-029..056)	in_progress (C1-C5 IMPLEMENTED 2026-05-28)	
commons/gitops promotion	Fixed (4a6b0ec)	Promoted pherald/internal/gitops → shared commons/gitops (L0, pure-stdlib: Runner/RepoRoot/FindScript/ParseUpstreams + TagExists/RemoteHasTag/LogSubjects) so all flavor modules reuse the git primitives (§11.4.74). No go.work/go.mod change (every flavor already requires commons).
§107 lazy-verifier fix (sherald+cherald)	Fixed (ed6c675,aef0207)	sherald + cherald main.go built the JWT verifier eagerly + os.Exit(1) before root.Execute() → EVERY subcommand (even version) died without HERALD_AUTH_MODE. The §43 CLI commands are for bare CI/cron invocation. Fixed by building the verifier LAZILY inside a flavor-local newServeCmd (cli.BindServeFlags+RunServe), mirroring pherald. serve still refuses anonymous. Binary-verified. See memory feedback_verify_subagent_cli_on_binary.
E53 Brotli flake	Fixed (dfca0be)	e2e E53 byte-compared two SEPARATE /v1/ safety_state responses, but the body carries live current_mem_percent/last_mem_sample_at/uptime_seconds → false FAIL. Replaced cross-call

ID	Status	What
§43 stragglers (HRD-034/047)	in_progress (IMPLEMENTED 2026-05-28, ba27a0e)	diff with a codec-agnostic <code>grep uptime_seconds</code> on the (decoded) body; kept the Content-Encoding:br policy check. §43 catalogue NOW COMPLETE — <code>sherald backup-snapshot</code> (§9.3 os.Link hardlinks → <code>.gate.recovered</code> , E121) + <code>sherald status-digest</code> (§11.4.45 Status.md sweep + <code>--apply Status_Summary regen</code> , E122). All 30 §43 commands (HRD-029..056) across <code>pherald/sherald/cherald/rherald/bherald/scherald</code> implemented + e2e-proven (E97-E122). commons_infra pgxTaskRepository — 16 of 17 methods implemented + REAL-PG verified. <code>GetByID/Update/Delete(soft) + UpdateStatus/Progress/Heartbeat/SaveCheckpoint + GetByStatus/GetPendingTasks/CountByStatus/GetTaskHistory + GetStaleTasks/GetByWorkerID + SaveResourceSnapshot/GetResourceSnapshots</code> ; migration <code>000013_task_resource_snapshots</code> (ledger 12→13). 14/14 integration round-trips PASS against live PG (<code>docs/qa/HRD-085-089-20260528T070000Z/integration_realpg.log</code>); e2e E123 (gated <code>HERALD_BATCHD_LIVE=1</code> — self-boot collides with the e2e's shared PG). The real-PG run caught + fixed a latent <code>LogEvent JSONB-as-[]byte→bytea 22P02</code> bug (<code>now string()</code>). <code>MoveToDeadLetter</code> still ErrUnsupported → HRD-090 (dead-letter table + retention, separate effort). PG boot: <code>podman machine start</code> (VM was never-started) + the integration suite self-boots <code>herald-postgres</code> via <code>QuickstartBoot</code> ; a stale <code>herald-e2e/herald-quickstart_herald-pg</code> volume caused a one-time 28P01 auth fail — removed both, recreated fresh with <code>herald_dev</code> . Generic messenger framework + Slack — entire Wave 7 (T1-T12) landed. Framework (HRD-110..114) Fixed pre-session: <code>channels.Channel</code> interface, <code>init()</code> -registered registry, per-channel inbox/., generalized <code>IsSelfEcho</code> via <code>SelfIdentity{Kind,Value}</code> , multi-channel <code>pherald listen</code> . This session: HRD-115 slack adapter <code>Socket Mode</code> (54d4316, vendored submodules/ slack-go@v0.16.0 /209ec09, 22 hermetic httpstest tests), HRD-116 qaherald raw-HTTP slack <code>MessengerClient</code> + channel-keyed Build (6811114, 30 tests, xoxb token NEVER in errors via <code>sanitizeError + TestSlack_Send_ErrorDoesNotLeakToken</code>), HRD-117 spec V3 r12→r13 (7f1b043, §11.0 + §32.2 LIVE + §43), HRD-118 e2e E124-E131 multi-channel invariants (17ecd0b, E127 LIVE-Slack triple-guarded against "no tests to run" PASS-bluff), HRD-119 paired §1.1 mutation gate 4 PASS / 0 FAIL (6737040, 3 mutations: blank- <code>IsSelfEcho</code> + drop-Register + drop-thread_ts), HRD-120 multi-channel §11.4.85 stress+chaos 12/12 PASS
Batch D (HRD-085..089)	in_progress (IMPLEMENTED 2026-05-28, 5899623)	
Wave 7 (HRD-110..121)	in_progress — CODE-COMplete 2026-05-28	

ID	Status	What
v0.6.0 tag	operator-gated	(86dfa1e, per-channel attribution + concurrency-under-fault), HRD-121 docs/guides/MESSENGER_CHANNELS.md operator guide (1037019, 610 lines + 3 siblings). HRDs stay in_progress pending operator-live Slack evidence (mirrors Wave 6's v0.4.0 deferred-to-T13b convention). Operator runs the live closed-loop against a real Slack workspace + commits docs/qa/HRD-115-LIVE-<run-id>/evidence per §107.x; then E127 converts SKIP→PASS automatically (its sc_anchor already cites the future live transcript path). Tag command: <code>git tag -a v0.6.0 -m "Wave 7 — multi-channel messenger framework"</code>. Required env for the live run: <code>HERALD_SLACK_BOT_TOKEN</code> (xoxb-) + <code>HERALD_SLACK_CHANNEL_ID</code> (Cxxx) + <code>HERALD_SLACK_APP_TOKEN</code> (xapp- for Socket Mode); operator-setup walkthrough at docs/guides/MESSENGER_CHANNELS.md §4 + §10 pre-deploy checklist. (1) Operator-live close-out wave — flip HRD-019..025 (Batch B bindings) + HRD-029..056 §43 (Batch C) + HRD-085..089 (Batch D) + HRD-115..121 (Wave 7) →Fixed once their respective live workspaces / PG instances / messenger workspaces are exercised. Bundles with v0.4.0 (Wave 6 deferred) + v0.6.0 (Wave 7) tags. (2) HRD-090 MoveToDeadLetter (dead-letter table + retention design) — the last ErrUnsupported TaskRepository method. (3) Wave 8 — design new channels (Max / Email / Discord / WhatsApp / Viber) on the now-stable Wave 7 framework + the supervised-restart redesign noted in HRD-120's T11 chaos doc-comment. (4) HRD-008 quickstart validation, HRD-015 gate-I8, HRD-081 (upstream containers podman/docker detect). (5) §11.4.93 SQLite (HRD-131, operator-deferred). Plan docs/superpowers/plans/2026-05-27-v1.0.0-execution.md + docs/superpowers/plans/2026-05-27-wave7-generic-messenger.md. HEAD=1037019; Wave 7 e2e = 142 PASS / 0 FAIL (T9 invariants green, T10 mutation gate 4/0, T11 stress+chaos 12/12); inheritance gate 15/0 post-.gitmodules slack-go-submodule add.
RESUME-AT (next units)	next	

§4. Next concrete steps

1. ~~**HRD-018 / 026 / 027 — DONE (v1.0.0 Batch A, b19cb43+854d301)**~~. The commons_constitution foundation (Evaluator interface, 12 event-class emit helpers, constitution_state+constitution_bindings migrations, bundle-hash captureer, mode-ladder) is complete + the constitution_audit write-through §107 bluff is closed + cherald admin mode-flip REST is wired; e2e 81/0. **Next §42 unit is Batch B (HRD-019..025, flavor constitution bindings)** — wire each flavor's per-binding

evaluation onto the now-complete foundation. Per §11.4.74, record Catalogue-Check: reuse|extend|no-match on each Batch B HRD before any Go code lands.

2. **HRD-008 quickstart validation** — On a fresh laptop with Podman or Docker:

```
git clone <Herald repo>
git submodule update --init
cd quickstart
podman build -t pherald:dev -f Dockerfile.pherald ../..
cp .env.example ../../.env && $EDITOR ../../.env
podman-compose -f docker-compose.quickstart.yml up -d
curl --retry 30 --retry-delay 2 http://localhost:24090/readyz
```

The current pherald serve returns "not implemented" — HRD-010/HRD-011/HRD-012 must land first to make the live curl POST /v1/events succeed end-to-end. Validation reveals which of the spec's assumptions (port ranges, Compose syntax, OTel collector config, Postgres healthcheck) hold against real infrastructure.

3. **HRD-010** — wire commons_storage/storage.go's MigrationDriver to golang-migrate; bring pgx + River + Redis client up; add integration tests under //go:build integration.
4. **HRD-011** — replace commons_messaging/channels/tgram/tgram.go stub with a live implementation against gopkg.in/telebot.v3 or github.com/mymmrac/telego; recorded HTTP fixtures under testdata/.
5. **HRD-012** — replace commons_messaging/dispatch/claude_code/claude_code.go's Dispatch stub with a real claude --resume invocation; capture session UUID; parse <<<HERALD-REPLY>>>.
6. **HRD-016** — scaffold pherald/internal/http/ with Gin routes per V3 §41; wire pherald serve to mount the Gin router on http_port.
7. **HRD-017** — propagate Universal §11.4.6X new mandates (spec-versioning + submodule-catalogue-first) into the constitution submodule.

§4a. Wave 6 live-test handoff (T10b — gating tag v0.4.0)

Operator-supplied credentials required — Wave 6's e2e invariants E63-E70 land as honest SKIPs until the live closed-loop runs against real Telegram + real Claude Code with real chat messages exchanged.

Step-by-step.

1. Export credentials in the shell that will drive the test:

```
export HERALD_TGRAM_BOT_TOKEN="<your-bot-token-from-BotFather>"
export HERALD_TGRAM_CHAT_ID="<numeric-chat-id-where-bot-is-admin>"
```

```
export HERALD_CLAUDE_BIN="$(command -v claude)" # path to
the claude CLI binary
export HERALD_PROJECT_NAME="Herald" # or any
operator-chosen session name
```

- Pick a <run-id> (timestamp + 4-char nonce):

```
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-$(uuidgen | cut -c1-4)"
QA_DIR="docs/qa/HRD-100-{$RUN_ID}"
mkdir -p "{$QA_DIR}/attachments"
```

- Type ONE message in the configured Telegram chat (script reads it via getUpdates within 60s).
- Run the live closed-loop test:

```
bash tests/test_wave6_live_loop.sh
```

The script builds pherald, observes the original message_id, starts pherald listen --bot-token "\$HERALD_TGRAM_BOT_TOKEN" --chat-id "\$HERALD_TGRAM_CHAT_ID" --qa-out-dir "\$QA_DIR" in the background, waits up to 45s for a reply with reply_to_message_id == original, and exits 0 on PASS.

- Capture supplementary logs:

```
cp /tmp/pherald-w6.log "{$QA_DIR}/pherald-listen.log"
# claude stdout/stderr — paths depend on the journaling
# setup. Copy whatever
# the test produced into {$QA_DIR}/claude-stdout.log +
# claude-stderr.log.
```

- Author a brief {\$QA_DIR}/README.md (5 lines minimum) narrating: who ran it, when, what message was sent, what reply came back, any anomalies. This is NOT auto-generated — operator-written narrative is the §107.x evidence anchor.
- git add {\$QA_DIR}/ && git commit -m "Wave 6 step 10b: docs/qa/HRD-100-{\$RUN_ID}/ live closed-loop evidence" then proceed to T13b (v0.4.0 tag + 4-mirror push).

If the script SKIPS (any env var unset): the test prints SKIP: <reason> and exits 0. That is honest §11.4.3 hardware-absent SKIP-with-reason, not a PASS. Tag v0.4.0 MUST NOT be created until at least one PASS run is committed to docs/qa/HRD-100-<run-id>/.

§4b. qaherald-auto resume runbook (next session — gating tag v0.5.1)

Status as of 2026-05-23 session end:

- HEAD 8ad4cb9 on main; all 4 mirrors in sync (github.com:vasic-digital/Herald.git, gitlab.com:vasic-digital/herald.git, gitflic.ru:vasic-digital/herald.git, gitverse.ru:vasic-digital/Herald.git).
- Tag v0.4.0 shipped (Wave 6 closed-loop). Tag v0.5.0 shipped (Wave 6.5 lifecycle implementations — premature: only S1+S2 live evidence + carries the pre-53a7ad3

envelope-action-guidance bug). Tag v0.5.1 is the target after qaherald-auto T6-T8 land + full 15-scenario evidence committed.

- Plan: docs/superpowers/plans/2026-05-23-qaherald-auto.md (commit 52ce679).
- Done: T1 skeleton + T2 messenger + T3 scenarios + T4 orchestrator + T5 report (combined commit 8ad4cb9 for T3-T5).
- TODO: **T6 pre-flight** + **T7 hermetic tests** + **T8 shell-adapter** then live run + tag.

Resume prerequisites

The following are **already provisioned** in ~/.zshrc (export-on-shell-start):

```
HERALD_TGRAM_BOT_TOKEN
    # @atmosphere_worker_bot — pherald-bot (Telegram bot
    # 8823384001)
HERALD_TGRAM_CHAT_ID      # -4946584787 (ATMOSphere Development
    # group)
HERALD_OPERATOR_IDS      # 2057253161 (the operator's user-id)
HERALD_QA_BOT_TOKEN      # @pherald_qa_bot — 2nd bot
    # (8971749017, privacy DISABLED, member of the group)
```

The qa-bot token is plaintext in ~/.zshrc — operator should /revoke via @BotFather after Wave 6.5 close-out and regenerate (the token leaked into one prior session's bash output earlier today, was never committed to git but lives in /private/tmp/claude-501/... operator-local files).

The containers herald-postgres (port 24100) and herald-redis (port 24200) are LIVE — both have been Up since the start of this session; data persists across podman restarts. Schema is at migration v12 (all 12 applied).

Resume sequence (suggested next-session order)

1. Pull state + verify:

```
cd ~/Projects/Herald && git pull origin main && git log --
    oneline -10
git status --short # expect clean
git stash list     # 2 stashes preserved: T8 Wave-5 salvage
    + premature S3-S5 transcript WIP
```

2. **Spawn T6 (pre-flight)** via a fresh subagent — read docs/superpowers/plans/2026-05-23-qaherald-auto.md Task 6 in full. Implementation lands at qaherald/internal/lifecycle/preflight.go + preflight_test.go. 10 gates: pherald-bot reachable via getMe, qa-bot reachable, qa-bot privacy disabled, qa-bot is group member, OPERATOR_IDS contains qa-bot, etc. Each gate has distinct exit code for diagnostics.

3. **Spawn T7 (full hermetic tests)** — extends T3-T5's 12 tests with httptest-based scenario simulation. Coverage targets: every scenario PASS path + every FAIL diagnostic + S9 SKIP path + S14 outbound-attachment sha256 round-trip + S11/S12/S13 inbound-attachment download + Issues.md/Fixed.md fs-mutation assertion.

4. **Spawn T8 (shell-script adapter)** — rewrites tests/
test_wave6.5_lifecycle.sh to delegate to qaherald lifecycle for
automated runs. Keeps the original --manual flag for the operator-typing interactive UX
as a fallback.

5. **Run the live e2e (the §107 watershed):**

```
cd ~/Projects/Herald
# Add qa-bot user-id to OPERATOR_IDS so S5/S6/S8/S10 succeed
export HERALD_OPERATOR_IDS="$
    {HERALD_OPERATOR_IDS},8971749017"
# Re-export from .zshrc (or just open a new shell)
source ~/.zshrc

# Build pherald + qaherald
go build -o /tmp/pherald ./pherald/cmd/pherald
go build -o /tmp/qaherald ./qaherald/cmd/qaherald

# Start pherald listen in background with QA journaling
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-w6.5live"
PHERALD_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`-pherald"
QAUTO_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`"
mkdir -p "${PHERALD_QA_DIR}/attachments" "${QAUTO_QA_DIR}/
    attachments"
/tmp/pherald listen --qa-out-dir "${PHERALD_QA_DIR}" --docs-
    dir docs &
PHERALD_PID=$!
trap 'kill -TERM $PHERALD_PID 2>/dev/null' EXIT
sleep 5

# Run qaherald lifecycle (T6-T8 must be done first)
/tmp/qaherald lifecycle \
    --pherald-bot-username=atmosphere_worker_bot \
    --pherald-qa-out-dir="${PHERALD_QA_DIR}" \
    --out="${QAUTO_QA_DIR}" \
    --run-id="${RUN_ID}"
# Exit 0 on all-PASS or all-PASS-with-S9-SKIP. Non-zero on
    any FAIL.
```

6. **Capture evidence:**

- `\${PHERALD\_QA\_DIR}/transcript.jsonl` — pherald's view (classifications, dispatch, replies)
- `\${QAUTO\_QA\_DIR}/transcript.jsonl` — qaherald-auto's view (sends, assertions)
- `\${QAUTO\_QA\_DIR}/report.md` — Markdown summary
- `\${QAUTO\_QA\_DIR}/attachments/<sha256>.<ext>` — content-addressed inbound + outbound attachments
- docs/Issues.md + docs/Fixed.md — mutated by S5/S6/S8/S10 (and reverted by S10 + S15 cleanups; both files end in valid state)

7. Author operator README + commit:

```
cat > "${QAUTO_QA_DIR}/README.md" <<EOF
# HRD-101 Wave 6.5 lifecycle evidence — <run-id>

<5+ lines narrating: who ran it, when, which scenarios
    PASSEd/FAILEd/SKIPped,
bot reply quality observations, any anomalies. Operator-
    written, NOT auto-generated.>
EOF
git add docs/qa/HRD-101-lifecycle-${RUN_ID}*
git commit -m "Wave 6.5 step 13: HRD-101 live 15-scenario
    lifecycle evidence (qaherald-auto)"
```

8. Tag v0.5.1 + 4-mirror push:

```
git tag -a v0.5.1 -m "Wave 6.5 close: full lifecycle live
    evidence + envelope action-guidance fix"
git push origin main
git push origin v0.5.1
# Verify all 4 mirrors converged on the same SHA + tag
for mirror in github.com:vasic-digital/Herald.git
    gitlab.com:vasic-digital/herald.git
    gitflic.ru:vasic-digital/herald.git
    gitverse.ru:vasic-digital/Herald.git; do
    git ls-remote "git@${mirror}" v0.5.1
done
```

Open / known constraints for tomorrow

- **S9 (non-operator Done: rejection):** requires HERALD_QA_BOT_TOKEN_NON_OPERATOR env (a THIRD bot account whose user-id is NOT in OPERATOR_IDS). Without it, S9 emits a SKIP-with-reason. Operator may register a third bot OR accept the SKIP (S9's logic is already unit-tested hermetically in T5's mutation gate).
- **First-scenario CC bootstrap:** ~30s on first inbound message because the Claude Code session must spawn. Subsequent scenarios use the cached session (~5-15s typical).
- **Issues.md mutation race:** if you run qaherald lifecycle while ANOTHER pherald listen is also processing in the same checkout, Issues.md may receive double-writes. Single-pherald-listen invocation is the safe default.
- **Postgres-SASL carry-over (RESOLVED 2026-05-27):** the e2e_bluff_hunt FAILs on Postgres-SASL invariants (E7-E12 + E14-E16 + E37-E42) are FIXED — 9b166c4 (auth env + PG password self-heal + Redis/tenant wiring + honest SKIP-with-reason) + bdb9f1 (Runner nil-Redis graceful PG-only fallback). At resolution e2e_bluff_hunt reported 60 PASS / 0 FAIL / 24 SKIP; the later GAP-3 session (d2a60d3) raised it to **78 PASS / 0 FAIL** with E81-E88 wired in.

Safe-shutdown checklist (HISTORICAL — end of 2026-05-23 session; superseded — current HEAD is d2a60d3)

Check	State
All committed work pushed to 4 mirrors	YES — HEAD 8ad4cb9 confirmed on github+gitlab+gitflic+gitverse
Local working tree clean	YES — <code>git status --short</code> empty
Stashes preserved	2 stashes: T8 Wave-5 salvage (older) + premature mid-test S3-S5 transcript WIP (newer, will be regenerated cleanly by qaherald-auto in next session)
Background pherald/qaherald processes	NONE running (<code>ps aux grep pherald</code> empty)
Container state	<code>herald-postgres + herald-redis</code> Up 17 hours — persist data across <code>podman stop && podman start</code> ; can be left running OR stopped (<code>podman stop herald-postgres herald-redis</code>); next session restart is <code>podman start herald-postgres herald-redis</code>
<code>/tmp/*.log</code> files	one operator-token leak shred-deleted earlier today; remaining transient logs contain no credentials (verified)
<code>.zshrc</code>	contains 4 <code>HERALD_*</code> exports including <code>HERALD_QA_BOT_TOKEN</code> plaintext — operator should <code>/revoke + regenerate</code> via @BotFather post-Wave-6.5 closure
Memory entries	All session findings persisted under <code>/Users/milosvasic/.claude/projects/-Users-milosvasic-Projects-Herald/memory/</code> — survive across sessions

Machine is safe to shut down. Resume tomorrow by `cd ~/Projects/Herald && git pull origin main` and start with qaherald-auto T6.

§5. Long-form pointers

- `docs/specs/mvp/specification.V3.md` — full active spec (Revision 4).
- `docs/specs/mvp/specification.V3.md#30-v2-self-review-log` — every review pass.
- `docs/guides/HERALD_CONSTITUTION.md` — §101..§106 extending Universal.
- `docs/guides/CONSTITUTION_INHERITANCE.md` — parent-discovery + gate.
- `tests/test_constitution_inheritance.sh` — the gate.
- `quickstart/` — HRD-008 scaffold.
- `commons/types.go` — the §11.0 type contract reference.